

Mohamed Abdallah Mohamed

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New Cairo, Egypt | Military status: Exempted

PROFILE

Software Engineering fresh graduate from The British University in Egypt / London South Bank University, graduating with Distinction (Honors) and a GPA of 3.7. Experienced in back-end development, full-stack web projects, AI-powered mobile applications, RAG-based systems, and AI-supported educational tools using Java, Spring Boot, Flutter, Firebase, Node.js, MongoDB, Qdrant, and LangChain.js. Delivered the freelance Konooz Studio inventory and point-of-sale platform, translating retail workflows and business requirements into a secure, tested back-end system. Currently gaining hands-on experience as a Software Engineer Intern at Celfocus, seeking opportunities as a Business Analyst or Back-End Developer, leveraging strong technical, analytical, and problem-solving skills.

EDUCATION

The British University in Egypt / London South Bank University
Dual Bachelor's Degree in Information and Computer Science (Software Engineering) | GPA: 3.7

Cairo, Egypt
2022 - 2026

Al Farouk Islamic Language School
Egyptian Diploma

Cairo, Egypt
2009 - 2022

Experience

Software Engineer Intern - Celfocus

Cairo, Egypt

Hybrid

July 2026 - Sep 2026

- Rotated across Back-End & Front-End Development, QA, DevOps, and AI, contributing to the Software Development Life Cycle (SDLC) on live projects.
- Developed and tested features using Java, Spring Boot, React, REST APIs, Git, and Docker, with focus on API integration, debugging, testing, and deployment.
- Analyzed requirements, identified technical issues, and collaborated with cross-functional teams to deliver solutions aligned with functional and business requirements.
- Gained hands-on experience with MCP, Prompt Engineering, and Agentic AI, exploring AI-driven automation and workflow optimization.

Freelance Software Engineer - Konooz Studio

Freelance Project

Back-End Development & Product Delivery

June 2026

- Translated dress-shop operations and stakeholder needs into functional workflows and business rules for model/colour/pack inventory, multi-device checkout, customer records, discounts, deposits, receipts, payments, refunds, and reporting.
- Engineered a server-authoritative REST API and data layer using TypeScript, Express.js, Prisma ORM, and PostgreSQL, with JWT authentication, Zod validation, transactional stock updates, stale-write protection, and immutable sale snapshots.
- Delivered and validated the production system using Docker, Cloudflare Workers/Pages, Neon PostgreSQL, GitHub Actions CI/CD, Vitest, Supertest, and Playwright; documented deployment, migration, validation, and rollback procedures.

Back-End Developer Intern - Connect Digital Solutions (CDS)

Cairo, Egypt

Hybrid

July 2025 - Sep 2025

- Developed back-end solutions using Java, Spring Boot, Spring Data JPA, Spring Security, and REST APIs, applying OOP, data structures, and clean code principles.
- Worked with microservices architecture, database integration, and API development to build scalable and maintainable back-end services.
- Applied requirements analysis, problem-solving, debugging, and software testing to support reliable application functionality.

CIB Online Internship

Cairo, Egypt

Cybersecurity and workplace readiness training

July 2024 - Aug 2024

- Completed training in Cybersecurity, Digital Transformation, Data Literacy, Financial Literacy, and Workplace Readiness, developing skills in risk awareness, data-driven decision-making, and digital business environments.
- Earned certificates in Emerging Talent for the Future Workplace (CIB), Human-Centric Interpersonal Skills (Frankfurt School), Financial Literacy & Entrepreneurial Skills (LinkedIn), and Digital Transformation & Data Literacy (SAS).

COURSES

Web Development using React.js (120 hrs.) - Information Technology Institute (ITI)

Online

Aug 2025

- Trained in React.js, JavaScript (ESNext), HTML5, Bootstrap, and client-side web development, with focus on component-based development and responsive web interfaces.

Introduction to Programming in Python - Harvard CS50

Online

June 2023 - July 2023

- Covered Python programming, OOP, data structures, algorithms, problem-solving, and file handling, strengthening programming and analytical skills.

PROJECTS

Digital Diet and Fitness Assistant (Flutter, Node.js, Firebase, Qdrant, LangChain.js)

Graduation Project - BUE

- Developed an AI-powered mobile application using Flutter and Node.js, based on a RAG architecture with Firebase authentication, a Qdrant vector database, LangChain.js orchestration, and LLM-based personalized diet and workout generation.

Intern CV Screening Platform (React, TypeScript, Vite, Axios, MSW, Storybook)

Celfocus Internship

- Developed a responsive CV-screening dashboard with protected authentication, REST API integration, candidate filtering, weighted evaluation reports, React Router, Atomic Design components, and Mock Service Worker.

Quality Management System (Java, Spring Boot, Spring Cloud, Spring Security, Microsoft SQL Server)

CDS Internship

- Translated business and functional requirements into secure RESTful microservices for quality evaluation, authentication, password recovery, user management, and RBAC using Spring Data JPA, Hibernate, JWT, MapStruct, Maven, JUnit, MockMvc, Git, and centralized Spring Cloud Config.

Agentic AI (Python, Tkinter, OpenCV, MediaPipe, FAISS)

Senior Year - BUE

- Implemented a multi-agent adaptive Python quiz system using Orchestrator, Perception, Adaptation, Hint, and Quiz Interface agents; integrated webcam gaze/emotion detection, FAISS hint retrieval, TinyLlama explanations, adaptive repetition, and Excel logging.

Hotel Management System (Java, MongoDB)

Senior Year - BUE

- Developed an object-oriented software project using Java and MongoDB, based on a client-server architecture with remote objects.

Mobile Application (Flutter, Firebase)

Senior Year - BUE

- Built a mobile application using Cloud Firestore and Firebase Authentication.

Full Stack Cinema Reservation Website (Tailwind CSS, Node.js, React, MongoDB, Express.js)

Third Year - BUE

- Built a MERN-stack website styled with Tailwind CSS, applying RESTful APIs, asynchronous programming, database modeling with Mongoose, and front-end/back-end integration.

Park Management System (Java, PHP)

Third Year - BUE

- Designed a desktop application using Java GUI and a PHP-based backend for data management; applied MVC and design patterns to strengthen software architecture understanding.

SKILLS

- **Programming Languages:** C, C++, C#, Python, Java, SQL, Dart, HTML, CSS, JavaScript, TypeScript, PHP
- **Frameworks and Technologies:** Flutter, Firebase, Node.js, Express.js, React.js, Tailwind CSS, MongoDB, PostgreSQL, Spring Boot, Spring Data JPA, Spring Security, Kafka, Microservices, Prisma ORM, Qdrant, LangChain.js, OpenCV, MediaPipe, FAISS, Git, Docker, Next.js, Vite, AWS, Cloudflare Workers/Pages, Neon, GitHub Actions
- **Software Engineering:** Requirements gathering and analysis, functional and non-functional requirements, process/workflow analysis, SDLC, OOP, data structures, algorithms, design patterns, database design, RESTful APIs, microservices, RBAC, Software Quality Assurance, DevOps, CI/CD, unit, integration, and end-to-end testing
- **AI:** CNNs, NLP, Model Training, RAG, Prompt Engineering, MCPs
- **Tools:** Microsoft Word, Microsoft Excel, Microsoft PowerPoint, Adobe Photoshop, VS Code, IntelliJ
- **Languages:** Arabic (native), English (fluent)